

SASEBO, Japan

WMO: 478120

Lat: 33.1597N

Lon: 129.7275E

Elev: 18

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.8	(c) 34.6	(d) 16.4	(e) 12.6	(f) 40.3	(g) 18.9	(h) 14.3	(i) 40.5	(j) 21.2	(k) 44.1	(l) 19.9	(m) 44.1	(n) 6.5	(o) 80	(p) 0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.7	(c) 91.8	(d) 77.9	(e) 89.9	(f) 77.9	(g) 88.1	(h) 77.4	(i) 80.3	(j) 87.0	(k) 79.6	(l) 86.0	(m) 78.9	(n) 85.1	(o) 9.2	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.5	83.8	77.8	144.6	83.4	77.1	140.9	83.0	44.0	87.3	43.1	86.2	42.4	85.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	17.6	15.3	29.3	95.6	1.8	1.9	28.0	96.9	27.0	98.0	26.0	99.1	24.7	100.5
			26.1	81.6	1.7	1.6	24.9	82.8	24.0	83.7	23.0	84.6	21.8	85.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	44.3	46.1	51.4	59.6	67.5	73.8	81.0	83.3	77.3	68.4	57.9	48.4
	DBStd	14.09	4.80	5.52	5.36	5.03	3.66	3.30	3.72	3.11	4.21	4.75	5.69	5.37
	HDD50	470	187	134	46	2	0	0	0	0	0	0	5	96
	HDD65	2537	643	529	420	171	17	0	0	0	0	21	221	515
	CDD50	5343	9	25	90	290	544	713	962	1032	820	569	242	47
	CDD65	1934	0	0	0	9	95	263	497	567	370	125	8	0
	CDH74	16506	0	0	0	17	300	1247	5058	6591	2855	432	6	0
	CDH80	5245	0	0	0	0	21	162	1653	2629	732	48	0	0
	WSAvg	6.4	6.7	6.7	7.1	6.8	6.1	5.8	6.5	6.4	6.3	6.1	5.9	6.6
	PrecAvg	79.6	2.8	3.4	5.1	6.6	7.3	13.0	12.8	9.5	7.8	3.9	4.4	3.1
(15) Precipitation	PrecMax	110.7	7.4	6.7	10.2	13.3	12.9	21.5	21.7	24.2	21.6	9.0	8.9	7.2
	PrecMin	42.5	0.7	0.9	2.0	2.8	2.6	1.4	1.6	1.2	2.6	0.2	0.5	0.2
	PrecStd	15.8	1.5	1.7	1.9	2.9	2.9	5.3	6.1	5.6	4.4	2.6	2.3	1.9
	PrecStd	15.8	1.5	1.7	1.9	2.9	2.9	5.3	6.1	5.6	4.4	2.6	2.3	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	65.5	69.5	76.4	82.7	86.2	93.9	95.4	90.8	84.5	74.9	65.2
		MCWB	56.3	58.3	58.8	63.1	66.2	72.2	78.5	78.6	76.3	71.3	64.6	58.2
	2%	DB	57.4	61.9	66.3	73.1	79.4	83.6	90.8	92.9	88.3	80.9	71.8	62.2
		MCWB	51.3	55.4	57.1	61.7	65.1	71.8	78.1	78.0	75.6	69.4	62.8	54.5
	5%	DB	54.8	58.3	63.5	70.8	77.3	81.8	88.8	90.9	86.2	78.3	69.3	59.7
		MCWB	48.4	52.0	55.9	60.6	65.0	72.0	78.1	78.1	74.7	67.5	61.5	52.4
	10%	DB	52.5	55.4	60.8	68.6	75.3	80.1	87.1	89.3	84.2	76.2	67.2	57.4
		MCWB	46.1	49.0	53.4	59.4	64.3	72.0	77.7	78.0	74.0	66.0	59.9	50.9
	0.4%	WB	57.8	60.7	62.8	68.8	73.0	77.8	81.7	81.6	79.6	75.9	68.1	61.4
		MCDB	60.4	63.1	65.8	72.0	76.3	82.0	87.5	88.8	85.0	80.8	70.9	63.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.2	57.1	59.8	65.7	70.6	76.7	80.3	80.6	78.3	72.9	65.6	56.6
		MCDB	55.9	60.7	63.8	69.7	74.1	80.5	86.2	88.0	84.2	78.3	69.2	60.1
	5%	WB	49.4	53.3	57.2	63.7	68.6	75.6	79.4	79.9	77.1	70.4	63.3	53.7
		MCDB	53.5	57.3	62.5	68.3	73.0	79.2	85.3	87.0	83.4	76.0	68.3	58.4
	10%	WB	46.8	49.6	54.7	61.4	66.9	74.1	78.8	79.1	76.0	67.8	60.8	51.4
		MCDB	51.9	54.4	60.0	66.7	72.5	77.9	84.6	86.0	82.2	74.0	66.2	56.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.3	13.1	13.6	13.5	10.7	9.4	10.7	11.2	12.5	12.7	11.8
		MCDBR	14.7	15.8	16.6	16.9	17.0	13.7	12.1	13.0	13.2	14.8	14.3	14.6
	5% WB	MCWBR	10.8	11.6	11.4	9.5	8.3	6.3	4.5	4.9	5.7	7.5	8.9	10.3
		MCDWR	13.1	13.3	14.0	12.2	11.3	8.2	10.1	10.9	10.6	11.3	11.8	12.7
	5% WB	MCWBR	11.6	12.3	12.0	9.4	8.1	5.2	4.5	4.8	5.4	7.4	9.2	10.9
		MCDWR	11.6	12.3	12.0	9.4	8.1	5.2	4.5	4.8	5.4	7.4	9.2	10.9
(40) Clear-Sky Solar Irradiance	taub		0.451	0.492	0.554	0.574	0.582	0.597	0.577	0.548	0.519	0.474	0.445	0.440
	taud		2.004	1.933	1.793	1.800	1.839	1.891	1.983	2.067	2.087	2.105	2.105	2.043
	Ebn at Noon		230	235	231	234	234	230	233	238	239	239	232	226
	Edh at Noon		45	53	66	69	66	63	57	52	49	44	40	41
(44) All-Sky Solar Radiation	RadAvg		679	921	1274	1538	1651	1410	1623	1689	1356	1134	815	632
	RadStd		65	95	87	129	193	191	234	159	136	113	81	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page